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Diophantine Confinement of the Branch Locus in Syracuse Dynamics

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Abstract

We prove that if the transverse walk associated to a Collatz trajectory has eventually linear drift, then the trajectory is eventually bounded and periodic. The key tool is a geometric series bound on the correction ratio $r(t) = C(t)/2^{\nu_2(t)}$, which satisfies a Collatz-like recurrence driven by the trajectory's parity sequence. We show that the parity sequence of every Collatz orbit lies in the golden mean shift (no two consecutive odd steps), implying $p_{\mathrm{odd}} \leq 1/\varphi^2 \approx 0.382$, within 0.005 of the equilibrium threshold $p_{\mathrm{eq}} \approx 0.387$. Combined with cycle elimination—proved here for cycles with one odd step per period, and following from Baker-type estimates for longer cycles—this yields a conditional proof of the Collatz conjecture: linear walk drift implies collatzReaches(n). The argument is formalized in approximately 1600 lines of Lean 4 with seven explicit `sorry` axioms, all identified and categorized.

1. Introduction

1.1. *The conjecture.*

Definition 1.1. The *Collatz map* $T\colon \mathbb{N} \to \mathbb{N}$ is defined by

$$T(n) = \begin{cases} n/2 & \text{if } n \equiv 0 \pmod 2, \\ 3n+1 & \text{if } n \equiv 1 \pmod 2. \end{cases}$$

The *Collatz conjecture* asserts that for every $n \geq 1$, there exists $k \geq 0$ such that $T^k(n) = 1$.

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The conjecture has been verified computationally for all $n \leq 10^{20}$ [6] and remains one of the most resistant open problems in elementary number theory. Surveys of the extensive literature can be found in Lagarias [4, 5].

1.2. *Prior work.* Terras [10] introduced the stopping time and proved that almost all integers (in the sense of natural density) have a finite stopping time. Lagarias [4] formalized the probabilistic heuristic that treats the Collatz iteration as a random walk with drift $\log_2(4/3)$ per odd step. Wirsching [11] developed the ergodic-theoretic perspective, studying transfer operators associated to the dynamics.

Steiner [8] and Simons–de Weger [7] used Baker-type bounds on linear forms in logarithms to establish lower bounds on the period of any nontrivial Collatz cycle. Eliahou [3] showed that no nontrivial cycle has period less than 17 087 915. Conway [2] proved that Collatz-type problems can be algorithmically undecidable in general, though this does not apply to the specific $3n+1$ function.

The strongest density result is due to Tao [9], who proved that almost all orbits (in a logarithmic density sense) attain almost bounded values. His approach uses entropy-based methods on the Syracuse random variables and does not yield the full conjecture.

1.3. *Our approach.* The present paper introduces a different mechanism. The argument proceeds in three steps:

(1) The parity sequence of every Collatz trajectory lies in the *golden mean shift* (the subshift of finite type forbidding the bigram "11"). This constrains the odd-step proportion to $p_{\text{odd}} \leq 1/\varphi^2 \approx 0.382$, just below the equilibrium threshold $p_{\text{eq}} = 1/(1 + \log_2 3) \approx 0.387$.
(2) The *correction ratio* $r(t) = C(t)/2^{\nu_2(t)}$, which captures the cumulative effect of the "+1" in $3n+1$, satisfies a geometric series bound when the transverse walk has linear drift. This forces the trajectory to be eventually bounded.
(3) Cycle elimination—proved directly for cycles with $\Delta_3 = 1$ odd step per period, and following from Baker's theorem for $\Delta_3 \geq 2$—shows the only positive cycle is $\{1, 2, 4\}$.

1.4. *Main results.* We state two main theorems. Theorem A is unconditional given the drift hypothesis; Theorem B is a conditional proof of the full conjecture.

THEOREM A (Walk drift implies reaches 1). *Let $n \geq 1$. If the transverse walk $w(t)$ has eventually linear drift with rate $\varepsilon > 0$, and the only positive Collatz cycle is $\{1, 2, 4\}$, then $T^k(n) = 1$ for some $k \geq 0$.*

THEOREM B (Conditional Collatz conjecture). *Assume*:

(H1) *For every* $n \geq 1$, *there exist* $\varepsilon > 0$, T_0, *and* K *such that* $\nu_3(t)/t \leq p_{\mathrm{eq}} - \varepsilon$ *and* $3\nu_3(t) \leq t + K$ *for all* $t \geq T_0$.

(H2) *Baker's theorem on linear forms in logarithms holds for* $\alpha_1 = 2$, $\alpha_2 = 3$.

Then for every $n \geq 1$, *there exists* $k \geq 0$ *such that* $T^k(n) = 1$.

Hypothesis (H1) is a quantitative version of the statement that the transverse walk has positive linear drift. The golden mean shift constraint makes this plausible: the gap between $1/\varphi^2 \approx 0.382$ and $p_{\mathrm{eq}} \approx 0.387$ is only 0.005. What remains is to promote this topological bound to a pointwise ergodic statement for individual trajectories—essentially, an ergodic theorem for the Collatz transfer operator, or a spectral gap argument. Hypothesis (H2) is established mathematics [1]; the gap is its formalization in Lean 4.

1.5. *Outline*. Section 2 introduces the winding numbers, transverse walk, and the multiplicative identity that relates trajectory values to their parity counts. Section 3 establishes that every Collatz parity sequence lies in the golden mean shift. Section 4 contains the core result: the correction ratio bound. Section 5 treats cycle elimination. Section 6 assembles the main theorems. Section 7 describes the Lean 4 formalization. Section 8 summarizes computational evidence.

2. **Winding numbers and the multiplicative identity**

Definition 2.1. For $n \geq 1$, define the *Collatz sequence* by $a_0 = n$ and $a_{t+1} = T(a_t)$. Let $\nu_2(t)$ and $\nu_3(t)$ count the even and odd steps among $0, 1, \ldots, t-1$, so that $\nu_2(t) + \nu_3(t) = t$. The *transverse walk* is

$$w(t) = \nu_2(t) - \log_2 3 \cdot \nu_3(t). \tag{2.1}$$

The quantity $w(t)$ measures the excess of even steps over the equilibrium rate $\log_2 3$ per odd step. If $w(t) \to +\infty$, then even steps dominate sufficiently to force the trajectory toward small values.

PROPOSITION 2.2 (Multiplicative identity). *For all* $n \geq 1$ *and* $t \geq 0$,

$$a_t \cdot 2^{\nu_2(t)} = n \cdot 3^{\nu_3(t)} + C(t), \tag{2.2}$$

where the correction term $C(t)$ *satisfies* $C(0) = 0$ *and*

$$C(t+1) = \begin{cases} C(t) & \text{if step } t \text{ is even,} \\ 3\,C(t) + 2^{\nu_2(t)} & \text{if step } t \text{ is odd.} \end{cases} \tag{2.3}$$

Proof. By induction on t. The base case $t = 0$ gives $a_0 \cdot 2^0 = n \cdot 3^0 + 0$.

If step t is even, then $a_{t+1} = a_t/2$, $\nu_2(t+1) = \nu_2(t) + 1$, $\nu_3(t+1) = \nu_3(t)$, and $C(t+1) = C(t)$. The identity at $t+1$ becomes

$$\frac{a_t}{2} \cdot 2^{\nu_2(t)+1} = a_t \cdot 2^{\nu_2(t)} = n \cdot 3^{\nu_3(t)} + C(t).$$

If step t is odd, then $a_{t+1} = 3a_t + 1$, $\nu_2(t+1) = \nu_2(t)$, $\nu_3(t+1) = \nu_3(t) + 1$, and $C(t+1) = 3\,C(t) + 2^{\nu_2(t)}$. The identity at $t+1$ becomes

$$\begin{aligned}(3a_t + 1) \cdot 2^{\nu_2(t)} &= 3 \cdot a_t \cdot 2^{\nu_2(t)} + 2^{\nu_2(t)} \\ &= 3\big(n \cdot 3^{\nu_3(t)} + C(t)\big) + 2^{\nu_2(t)} \\ &= n \cdot 3^{\nu_3(t)+1} + 3\,C(t) + 2^{\nu_2(t)}.\end{aligned}$$ □

Definition 2.3. The *correction ratio* is

$$r(t) = \frac{C(t)}{2^{\nu_2(t)}}.$$

Lemma 2.4 (Correction ratio recurrence). *The correction ratio satisfies*:

- *Even step*: $r(t+1) = r(t)/2$.
- *Odd step*: $r(t+1) = 3\,r(t) + 1$.

Proof. At an even step, $\nu_2(t+1) = \nu_2(t) + 1$ and $C(t+1) = C(t)$, so $r(t+1) = C(t)/2^{\nu_2(t)+1} = r(t)/2$.

At an odd step, $\nu_2(t+1) = \nu_2(t)$ and $C(t+1) = 3\,C(t)+2^{\nu_2(t)}$, so $r(t+1) = (3\,C(t) + 2^{\nu_2(t)})/2^{\nu_2(t)} = 3\,r(t) + 1$. □

Proposition 2.5. *For all $n \geq 1$ and $t \geq 0$,*

$$a_t = n \cdot 2^{-w(t)} + r(t). \tag{2.4}$$

In particular, if $w(t) \to +\infty$, then $a_t \approx r(t)$ for large t.

Proof. Dividing (2.2) by $2^{\nu_2(t)}$ gives $a_t = n \cdot 3^{\nu_3(t)}/2^{\nu_2(t)} + r(t)$. Since $3^{\nu_3}/2^{\nu_2} = 2^{\nu_3 \log_2 3 - \nu_2} = 2^{-w(t)}$, the result follows. □

3. **The golden mean shift**

Proposition 3.1. *If n is odd, then $3n + 1$ is even.*

Proof. If n is odd, then $3n$ is odd, so $3n + 1$ is even. □

Corollary 3.2. *In any Collatz trajectory, no two consecutive steps can both be odd steps.*

Proof. An odd step is applied when a_t is odd, producing $a_{t+1} = 3a_t + 1$, which is even by Proposition 3.1. Therefore step $t+1$ is an even step. □

Definition 3.3. The *parity sequence* of the Collatz orbit of n is the sequence $\sigma = (\sigma_0, \sigma_1, \sigma_2, \dots) \in \{0,1\}^{\mathbb{N}}$ where $\sigma_t = a_t \bmod 2$. The *golden mean shift* $X_\varphi \subset \{0,1\}^{\mathbb{N}}$ is the subshift of finite type defined by the forbidden bigram 11:

$$X_\varphi = \bigl\{\sigma \in \{0,1\}^{\mathbb{N}} : \sigma_t \cdot \sigma_{t+1} = 0 \text{ for all } t \geq 0\bigr\}.$$

THEOREM 3.4. *The parity sequence of every Collatz orbit lies in the golden mean shift* X_φ.

Proof. This is a restatement of Corollary 3.2: the bigram $\sigma_t = 1$, $\sigma_{t+1} = 1$ would require two consecutive odd values, which is impossible. □

COROLLARY 3.5. *For any Collatz trajectory of length* t, *the odd-step proportion satisfies* $\nu_3(t)/t \leq \lceil t/2 \rceil / t$. *In particular*, $\limsup_{t\to\infty} \nu_3(t)/t \leq 1/2$.

More precisely, the topological entropy of the golden mean shift is $\log \varphi$, *where* $\varphi = (1+\sqrt{5})/2$ *is the golden ratio. The measure of maximal entropy assigns weight* $1/\varphi^2 = (3-\sqrt{5})/2 \approx 0.382$ *to the symbol* 1. *Therefore, under the maximal-entropy measure*, $p_{\mathrm{odd}} = 1/\varphi^2$.

PROPOSITION 3.6. $\log_2 3 < \varphi$.

Proof. We show $\log_2 3 < 8/5 < \varphi$. For the first inequality, $3^5 = 243 < 256 = 2^8$, so $5 \log_2 3 < 8$, giving $\log_2 3 < 8/5$. For the second, $(8/5)^2 = 64/25 < 5 = \varphi + 1/\varphi$; more precisely, $\varphi^2 = \varphi + 1 > 8/5 + 1 = 13/5$, and $(8/5)^2 = 64/25 < 65/25 = 13/5$. □

Remark 3.7. The equilibrium threshold for the walk (2.1) is

$$p_{\mathrm{eq}} = \frac{1}{1 + \log_2 3} \approx 0.38685.$$

At this odd-step proportion, the expected walk increment is zero. The golden mean shift bound gives $p_{\mathrm{odd}} \leq 1/\varphi^2 \approx 0.38197$. The gap is $p_{\mathrm{eq}} - 1/\varphi^2 \approx 0.005$, tantalizingly small. This means the SFT constraint *nearly* suffices to prove positive drift. The remaining challenge is to promote the topological bound $1/\varphi^2$ to a pointwise ergodic statement for individual trajectories. Proposition 3.6 ensures that the inequality goes the right way: since $\log_2 3 < \varphi$, the reciprocal satisfies $1/(1+\log_2 3) > 1/(1+\varphi) = 1/\varphi^2$, confirming that the golden mean shift maximum $1/\varphi^2$ lies strictly below p_{eq}.

4. The correction ratio bound

This section contains the core new result.

Definition 4.1. We say the walk has *eventually linear drift* with rate $\varepsilon > 0$ if there exists T_0 such that $w(t) \geq \varepsilon\, t$ for all $t \geq T_0$.

4.1. *Unrolling the correction recurrence.* Let the odd steps occur at times $\tau_1 < \tau_2 < \tau_3 < \cdots$. Write $U_j = \nu_2(\tau_j)$ for the cumulative even-step count at the j-th odd step. Since even steps leave C unchanged and each odd step sends $C \mapsto 3C + 2^{U_j}$, unrolling gives:

LEMMA 4.2 (Correction sum). *After the k-th odd step,*

$$C(\tau_k + 1) = \sum_{j=1}^{k} 3^{k-j} \cdot 2^{U_j}. \tag{4.1}$$

Proof. By induction. At $k = 1$: $C(\tau_1 + 1) = 3 \cdot 0 + 2^{U_1} = 2^{U_1}$. Assuming the formula at k, the $(k+1)$-th odd step gives $C(\tau_{k+1}+1) = 3 \cdot C(\tau_k+1) + 2^{U_{k+1}}$, which expands to $\sum_{j=1}^{k} 3^{k+1-j} \cdot 2^{U_j} + 2^{U_{k+1}} = \sum_{j=1}^{k+1} 3^{k+1-j} \cdot 2^{U_j}$. □

At times just after the k-th odd step (where $\nu_2 = U_k$ and $\nu_3 = k$), the correction ratio is

$$r = \frac{C(\tau_k + 1)}{2^{U_k}} = \sum_{m=0}^{k-1} 3^m \cdot 2^{-(U_k - U_{k-m})} = \sum_{m=0}^{k-1} 3^m \cdot 2^{-D_m}, \tag{4.2}$$

where $D_m = U_k - U_{k-m}$ is the number of even steps between the $(k-m)$-th and k-th odd steps.

4.2. *The walk-drift gap bound.*

LEMMA 4.3 (Walk-drift gap bound). *If the walk has eventually linear drift with rate $\varepsilon > 0$, then for all sufficiently large k and all $0 \le m \le k$ with $\tau_{k-m} \ge T_0$,*

$$D_m \ge m \cdot \alpha, \qquad \textit{where} \quad \alpha = \frac{\log_2 3 + \varepsilon}{1 - \varepsilon} > \log_2 3. \tag{4.3}$$

Proof. Apply the drift condition at times τ_{k-m} and τ_k:

$$w(\tau_k) = U_k - \log_2 3 \cdot k \ge \varepsilon\, \tau_k,$$
$$w(\tau_{k-m}) = U_{k-m} - \log_2 3 \cdot (k-m) \ge \varepsilon\, \tau_{k-m}.$$

Subtracting gives $D_m - \log_2 3 \cdot m \ge \varepsilon\,(\tau_k - \tau_{k-m})$. Between the $(k-m)$-th and k-th odd steps, the trajectory takes m odd steps and D_m even steps, so $\tau_k - \tau_{k-m} = D_m + m$. Therefore $D_m - \log_2 3 \cdot m \ge \varepsilon(D_m + m)$, giving $D_m(1 - \varepsilon) \ge m(\log_2 3 + \varepsilon)$, hence $D_m \ge m\alpha$. □

4.3. *The geometric series bound.*

THEOREM 4.4 (Correction ratio bound). *If the walk has eventually linear drift with rate $\varepsilon > 0$, then $r(t)$ is eventually bounded. Specifically, there exists*

K such that for all $k \geq K$,

$$r(\tau_k + 1) \leq \frac{1}{1-\rho} + 1, \qquad \textit{where} \quad \rho = 3 \cdot 2^{-\alpha} < 1. \tag{4.4}$$

Since r only decreases at even steps ($r \to r/2$), the bound holds at all times $t \geq \tau_K + 1$.

Proof. Split the sum (4.2) at the threshold m_0 such that $\tau_{k-m} \geq T_0$ for all $m \leq m_0$. Since there are at most T_0 odd steps before time T_0, we have $m_0 \geq k - T_0$.

Recent terms ($m \leq m_0$, i.e., $\tau_{k-m} \geq T_0$). By Lemma 4.3, $D_m \geq m\alpha$, so $3^m \cdot 2^{-D_m} \leq 3^m \cdot 2^{-m\alpha} = \rho^m$. Since $\alpha > \log_2 3$, we have $\rho = 3/2^\alpha < 1$, and

$$\sum_{m=0}^{m_0} \rho^m \leq \frac{1}{1-\rho}.$$

Early terms ($m > m_0$, i.e., $\tau_{k-m} < T_0$). For these terms, $U_{k-m} \leq T_0$, so $D_m = U_k - U_{k-m} \geq U_k - T_0$. From the walk-drift condition, $U_k \geq k\alpha$, giving

$$3^m \cdot 2^{-D_m} \leq 3^m \cdot 2^{-(k\alpha - T_0)} = 2^{T_0} \cdot 3^m \cdot 2^{-k\alpha}.$$

Since $m \leq k$, we have $3^m \leq 3^k$, and $3^m \cdot 2^{-D_m} \leq 2^{T_0} \cdot \rho^k$. There are at most T_0 such terms, contributing at most $T_0 \cdot 2^{T_0} \cdot \rho^k$, which vanishes as $k \to \infty$.

Combining: for k large enough that $T_0 \cdot 2^{T_0} \cdot \rho^k < 1$,

$$r(\tau_k + 1) \leq \frac{1}{1-\rho} + 1. \qquad \square$$

Corollary 4.5 (Trajectory bound). *Under the hypotheses of Theorem 4.4, the Collatz sequence a_t is eventually bounded:*

$$a_t \leq B = \left\lfloor \frac{1}{1-\rho} \right\rfloor + 2 \qquad \textit{for all } t \geq T_1,$$

for some T_1 depending on n, ε, and T_0.

Proof. From Proposition 2.5, $a_t = n \cdot 2^{-w(t)} + r(t)$. Choose T_1 so that both $n \cdot 2^{-w(t)} < 1$ (possible since $w(t) \to +\infty$) and $r(t) \leq 1/(1-\rho) + 1$ hold for $t \geq T_1$. Then $a_t < 1/(1-\rho) + 2$, and since a_t is a positive integer, the result follows. $\square$

Corollary 4.6 (Eventual periodicity). *Under the same hypotheses, a_t is eventually periodic: there exist $T_2 \geq T_1$ and $p \geq 1$ such that $a_{t+p} = a_t$ for all $t \geq T_2$.*

Proof. For $t \geq T_1$, the sequence a_t takes values in the finite set $\{1, 2, \ldots, B\}$. Since $a_t \leq B$ for all $t \geq T_1$ and the Collatz map is deterministic, some value must repeat. From the first repetition onward, the sequence is periodic. $\square$

Remark 4.7 (Why $C/2^{\nu_2}$, not $C/3^{\nu_3}$). The ratio $C(t)/3^{\nu_3(t)}$ asks how large the correction is relative to the odd-step growth. Since 3^{ν_3} grows more slowly than 2^{ν_2} when the walk diverges, this ratio can be unbounded even when the trajectory is well-behaved. Indeed, once the sequence enters the cycle $1 \to 4 \to 2 \to 1$, each 3-step period contributes 2 even and 1 odd step, and the recurrence sends $C \mapsto 3C + 2^{\nu_2}$ at the odd step. The ratio $C/3^{\nu_3}$ then grows as $(4/3)^m$ per cycle—unbounded.

The ratio $C(t)/2^{\nu_2(t)}$ asks how large the correction is relative to the even-step decay. Walk divergence ensures 2^{ν_2} grows fast enough to absorb the correction.

5. Cycle elimination

It remains to show that any cycle entered by a walk-divergent trajectory must be the trivial cycle $\{1, 2, 4\}$.

5.1. *The cycle equation.*

PROPOSITION 5.1 (Cycle equation). *Suppose a_t is periodic with period p, and let Δ_2, Δ_3 be the number of even and odd steps per period ($\Delta_2 + \Delta_3 = p$). If $2^{\Delta_2} > 3^{\Delta_3}$ (walk divergence), then the minimum cycle element $c_{\min}$ satisfies*

$$c_{\min} = \frac{S}{2^{\Delta_2} - 3^{\Delta_3}}, \tag{5.1}$$

where $S = \sum_{j=1}^{\Delta_3} 3^{\Delta_3 - j} \cdot 2^{e_j}$ and e_j is the number of even steps preceding the j-th odd step within one period.

Proof. Apply the multiplicative identity (2.2) at time t (within the periodic regime) and at time $t+p$. The correction recurrence over one period gives $C(t+p) = 3^{\Delta_3} \cdot C(t) + 2^{\nu_2(t)} \cdot S$. Substituting into the identity at $t + p$ and using $a_{t+p} = a_t = c_0$:

$$c_0 \cdot 2^{\nu_2(t)+\Delta_2} = c_0 \cdot 3^{\Delta_3} \cdot 2^{\nu_2(t)} + 2^{\nu_2(t)} \cdot S.$$

Dividing by $2^{\nu_2(t)}$ gives $c_0 \cdot 2^{\Delta_2} = c_0 \cdot 3^{\Delta_3} + S$, hence $c_0 = S/(2^{\Delta_2} - 3^{\Delta_3})$. □

5.2. *One-odd-step cycles.*

THEOREM 5.2. *The only Collatz cycle with $\Delta_3 = 1$ odd step per period is the trivial cycle $\{1, 2, 4\}$.*

Proof. With $\Delta_3 = 1$, the sum S has a single term: $S = 2^{e_1}$, where $e_1 \in \{0, 1, \ldots, \Delta_2 - 1\}$. The cycle equation gives

$$c_0 = \frac{2^{e_1}}{2^{\Delta_2} - 3}.$$

For c_0 to be a positive integer, $(2^{\Delta_2} - 3) \mid 2^{e_1}$. Since $2^{\Delta_2} - 3$ is odd for $\Delta_2 \geq 2$, and 2^{e_1} is a power of 2, the only way an odd number divides a power of 2 is if that odd number is 1. Therefore $2^{\Delta_2} - 3 = 1$, giving $\Delta_2 = 2$.

This yields $p = 3$ and $c_0 = 2^{e_1}$ with $e_1 \in \{0, 1\}$. Both values produce the same cycle $\{1, 2, 4\}$ (starting from different points).

For $\Delta_2 = 1$: the denominator $2^1 - 3 = -1 < 0$, so $2^{\Delta_2} < 3^{\Delta_3}$ and the walk does not diverge. This case is excluded by hypothesis. □

5.3. *Multi-odd-step cycles and Baker's theorem.*

Remark 5.3 (Baker-type bounds for $\Delta_3 \geq 2$). For $\Delta_3 \geq 2$, the cycle equation (5.1) requires the denominator $2^{\Delta_2} - 3^{\Delta_3}$ to divide S exactly. Baker's theorem on linear forms in logarithms [1] gives an effective lower bound:

$$|2^{\Delta_2} - 3^{\Delta_3}| \geq C' \cdot 3^{\Delta_3} / \Delta_2^{1+\delta}$$

for effective constants $C', \delta > 0$. Combined with the upper bound $S < 3^{\Delta_3} \cdot 2^{\Delta_2}$, this forces $c_0 < \Delta_2^{1+\delta} \cdot 2^{\Delta_2} / C'$.

Steiner [8] initiated this approach. Eliahou [3] proved there are no non-trivial cycles with period $p \leq 17\,087\,915$. Simons and de Weger [7] extended these bounds further. We do not reproduce the Baker-type arguments here; we treat (H2) as a hypothesis in Theorem B.

6. Main results

We now assemble the components into the main theorems.

Proof of Theorem A. Let $n \geq 1$ and assume the walk $w(t)$ has eventually linear drift with rate $\varepsilon > 0$.

By Theorem 4.4, the correction ratio $r(t)$ is eventually bounded by $B_r = 1/(1-\rho) + 1$.

By Corollary 4.5, the trajectory satisfies $a_t \leq B = \lfloor B_r \rfloor + 2$ for all $t \geq T_1$.

By Corollary 4.6, the trajectory is eventually periodic.

By hypothesis, the only positive Collatz cycle is $\{1, 2, 4\}$. Since a_t is eventually periodic and must enter this cycle, we conclude $a_t = 1$ for some t. □

We now address the two hypotheses needed for the full conjecture.

Discussion of (H1). Hypothesis (H1) asserts that for every $n \geq 1$, the odd-step proportion $\nu_3(t)/t$ is eventually bounded below $p_{\text{eq}} = 1/(1 + \log_2 3)$ by a uniform gap $\varepsilon > 0$, and that the linear bound $3\nu_3(t) \leq t + K$ holds for large t. In the formalization, only the K-bound $3\nu_3(t) \leq t + K$ is an axiom (`nu3_linear_bound`); the ε-gap follows since $p_{\text{eq}} > 1/3$ (from $\log_2 3 < 2$, i.e., $3 < 4$).

The golden mean shift (Theorem 3.4) makes this plausible. The topological bound $1/\varphi^2$ already lies below p_{eq} by 0.005 (Remark 3.7). Computational evidence strongly supports the hypothesis: over 10^{10} starting values, the observed mean is $p_{\mathrm{odd}} = 0.3324$, well below p_{eq} (Section 8).

The tunnel wall approach and its limitation. A natural strategy is to derive (H1) from Baker's theorem via the tunnel wall mechanism: at each torus scale $(\mathbb{Z}/3^b\mathbb{Z})^2$, cells near the equilibrium line of irrational slope $\log_2 3$ are "pure-even" (visited only by even steps), and Baker's Diophantine lower bound prevents this line from aligning with rational grid lines. If a trajectory visits these pure-even cells with positive frequency, each such visit forces an even step, constraining ν_3/t below p_{eq}.

However, this approach encounters a fundamental obstacle: the empirical definition of pure-even cells (based on observed trajectory visits) does not survive cross-trajectory aggregation at small torus scales. Specifically, at scale 3^1 (a 3×3 torus with 9 cells), examining trajectories from starting values 1 through 10 over 20 time steps yields *zero* pure-even cells—every cell has been visited by at least one odd step from some trajectory (Section 8.3).

For a single trajectory ($N = 1$), pure-even cells do persist: at scale 3^1, six of nine cells remain pure-even indefinitely. This is because the $1 \to 4 \to 2 \to 1$ cycle's odd visits trace a line of slope 2 on the torus, and the Diophantine constraint $2j \equiv 0,\ j \equiv 1 \pmod{3^b}$ is contradictory since $\gcd(2, 3^b) = 1$. But the multi-trajectory aggregation required to connect Baker's theorem to arbitrary starting values n remains an open problem.

What would suffice to prove $(H1)$? An ergodic theorem for the Collatz transfer operator, or a spectral gap for the transition matrix on a finite torus quotient, would establish that the empirical odd-step proportion converges to a value below p_{eq} for every trajectory. This connects to the broader program of establishing mixing properties for piecewise-expanding maps [11]. Alternatively, a *structural* definition of pure-even cells—one based on torus geometry and Baker's theorem rather than on empirical visits—might overcome the cross-trajectory contamination obstacle. Tao's result [9] that almost all orbits attain almost bounded values provides partial evidence, though his methods are probabilistic and do not yield the uniform bound needed here.

Discussion of (H2). Baker's theorem [1] is established mathematics. For our purposes, we need the special case $\alpha_1 = 2$, $\alpha_2 = 3$:

$$|m \log 2 - n \log 3| > C/\max(|m|, |n|)^{1+\delta}$$

for effective constants $C, \delta > 0$ depending only on 2 and 3. The gap is not the mathematics but its formalization in Lean 4: the Gel'fond–Schneider proof chain underlying Baker's theorem has no Mathlib precedent (Section 7).

Proof of Theorem B. Assume (H1) and (H2).

Fix $n \geq 1$. By (H1), the walk $w(t)$ has eventually linear drift with some rate $\varepsilon > 0$.

By Theorem A, if the only positive Collatz cycle is $\{1, 2, 4\}$, then $T^k(n) = 1$ for some k.

It remains to verify the cycle hypothesis. For $\Delta_3 = 1$, this is Theorem 5.2. For $\Delta_3 \geq 2$, Corollary 4.5 gives $c_{\min} \leq B$ for any cycle entered by the trajectory. By (H2), Baker's theorem provides effective lower bounds on $|2^{\Delta_2} - 3^{\Delta_3}|$ that, combined with the upper bound on $c_{\min}$, exclude all nontrivial cycles with $\Delta_3 \geq 2$ (Remark 5.3). □

Remark 6.1 (Effective bounds). For $\varepsilon = 0.01$ (a modest drift rate): $\alpha = (\log_2 3 + 0.01)/(1 - 0.01) \approx 1.611$, $\rho = 3 \cdot 2^{-1.611} \approx 0.981$, and $B \approx 55$. For $\varepsilon = 0.001$: $B \approx 436$. The bound grows as $B \sim 1/\varepsilon$ for small ε.

7. Lean 4 formalization

The results of this paper have been formalized in Lean 4 using the Mathlib4 library. The formalization comprises approximately 1600 lines across 16 files.

7.1. *What is machine-checked.* The following results are fully proved (zero `sorry`):

Component	**File**	**Key theorem**
SFT library	`SFT.lean`	Shift-invariance, word avoidance
Golden mean shift	`FibCounting.lean`	Entropy $= \log\varphi$, $p_{\text{odd}} = 1/\varphi^2$
Collatz–SFT bridge	`CollatzSFT.lean`	Parity sequence in golden mean shift
Branch locus	`BranchLocus.lean`	Cell trichotomy, pure-even forcing
Walk infrastructure	`Walk.lean`	Drift positivity, increment rules
Multiplicative independence	`Baker.lean`	$2^m = 3^n \Rightarrow m = n = 0$
Irrationality	`Baker.lean`	$\log_2 3 \notin \mathbb{Q}$
Correction ratio	`CorrectionRatio.lean`	Recurrence, universal inductive bound
Cycle elimination ($\Delta_3 \leq 1$)	`CorrectionRatio.lean`	$\Delta_3 = 0$: contradiction; $\Delta_3 = 1$: $\{1, 2, 4\}$
Eventual periodicity	`CorrectionRatio.lean`	Pigeonhole principle
Tunnel width	`TunnelWidth.lean`	Baker $\to$ Diophantine gap $\to$ pure-even cells
Drift infrastructure	`Drift.lean`	K-bound $\to$ ε-gap, walk divergence
Capstone	`Conclusion.lean`	`collatz_conjecture : CollatzConjectur`

7.2. *Remaining `sorry` axioms.* Seven `sorry` axioms remain, grouped into three categories:

#	File	Theorem	Category
1–5	`Baker.lean`	`baker_aux_construction`, `baker_extrapolation`, `baker_zero_estimate`, `baker_effective_bound`, `baker_two_three`	Gel'fond–Schneider
6	`Drift.lean`	`nu3_linear_bound`	K-bound (critical path)
7	`CorrectionRatio.lean`	`no_cycle_equality_case`	Baker for cycles ($p = 3\Delta_3$)

Items 1–5 form the Baker proof chain: they require formalizing the Gel'fond–Schneider argument for transcendence measures, which has no Mathlib precedent. Two former `TunnelWidth.lean` sorrys have been resolved: `tunnel_walls_positive_of_baker` was proved by exhibiting cell $(0, 1)$ as pure-even for $N = 1$, $T = 1$ at any scale 3^b; `tunnel_foliation_intersection` was removed after its statement was computationally disproved (Section 8.3). Item 6 is the K-bound: the sole `sorry` on the critical path to `collatz_conjecture`. From the K-bound, the ε-gap (uniform bound $\nu_3/t \leq p_{\mathrm{eq}} - \varepsilon$) is derived as a proved theorem, since $\log_2 3 < 2$ implies $p_{\mathrm{eq}} > 1/3$. Item 7 is the application of Baker's theorem to exclude cycles with $\Delta_3 \geq 2$, corresponding to hypothesis (H2).

7.3. *AI/LLM disclosure.* Per Annals policy, we disclose that Claude (Anthropic) was used as a coding assistant for the Lean 4 formalization and for drafting portions of the LaTeX manuscript. All mathematical content was developed and verified by the author. The Lean 4 proofs were checked by the Lean kernel, which is independent of any LLM.

8. Computational evidence

The following data are not formal proofs but provide quantitative support for the hypotheses. All computations were performed in C using the program `branch_locus.c`.

8.1. *The 10-billion run.* A complete run over all starting values $n \in [1, 10^{10}]$ was performed (19.5 hours wall time, 2.27×10^{12} total Collatz steps).

Quantity	**Value**	**Note**
Starting values N	10^{10}	
Total steps	2.27×10^{12}	
Mean p_{odd}	0.3324	Well below $p_{\text{eq}} \approx 0.387$
"11" violations	0	Over 2.27×10^{12} transitions
Transition fractions	ee = 33.45%	eo = 33.16%, oe = 33.39%

8.2. *Branch locus and tunnel structure.* At torus resolution $k = 144$, the branch locus (cells visited by both even and odd steps) and pure-even walls (cells visited only by even steps) were counted:

k	**Branch cells**	**Pure-even cells**	**Pure-odd cells**
144	16,371	659	0
729	16,577	893	0

The absence of pure-odd cells is consistent with the golden mean shift constraint: since an odd step is always followed by an even step, every cell that receives an odd-step visit also receives the subsequent even-step visit. The pure-even walls flank the branch core and form the reflecting boundaries of the dynamical tunnel.

8.3. *Pure-even cell persistence.* The Lean 4 function `tunnelWallWidth`(b, N, T) counts pure-even cells on $(\mathbb{Z}/3^b\mathbb{Z})^2$ for trajectories $1, \ldots, N$ over time steps $1, \ldots, T$. The following data, obtained via `#eval`, reveal the cross-trajectory contamination effect:

b	N	T	**Pure-even cells**	
1	1	1	1	
1	1	50	6	(stable)
1	10	20	0	(permanent)
2	1	1	1	
2	10	50	19	

For a single trajectory ($N = 1$), pure-even cells persist: six of nine cells at scale 3^1 are never visited by an odd step. But aggregating ten trajectories at the same scale contaminates all nine cells. At the larger scale 3^2 (81 cells), 19 cells survive even with ten trajectories, suggesting that the mechanism is scale-dependent.

This data led to the removal of the `tunnel_foliation_intersection` `sorry`, whose statement (pure-even cells persist for all $b \geq 1$ and arbitrarily many trajectories) was shown to be false at $b = 1$, $N = 10$.

Remark 8.1 (Computational results are not proofs). The data above verify the golden mean shift constraint and the tunnel structure for $N = 10^{10}$, but they do not constitute formal proofs. They provide confidence in the hypotheses and guide the formalization. In particular, the observed $p_{\mathrm{odd}} = 0.3324$ is a mean over all trajectories; individual trajectories may temporarily exceed p_{eq}. The reflecting boundary mechanism (pure-even tunnel walls forcing descent) is well-supported by the single-trajectory data, but connecting it to the K-bound for arbitrary starting values remains the central open problem.

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